DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**CENTRAL LINE INSERTION**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-PICU-007 (v3)**		
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1.0 PURPOSE

1.1. To unify the procedure in assisting central line insertion procedure.

1.2. To deliver hyperosmolar solutions, long term antibiotics and TPN.

1.3. To measure CVP

1.4. To prevent multiple venipuncture

2.0 DEFINITION/ABBREVIATION

2.1. **CVP** - Central Venous Pressure.

2.2. **TPN** - Total Parenteral Nutrition.

2.3. **Central venous catheter** a thin, flexible tube that is inserted into a vein in the upper arm, thigh, or neck or below the collarbone. It is guided (threaded) into a large vein near the heart called the vena cava or into the right atrium of the heart. It is used to give intravenous fluids, blood transfusions, and chemotherapy and other drugs, and for taking blood samples. It avoids the need for repeated needle sticks.

3.0 RESPONSIBILITY

3.1 **Physician** is responsible for performing the procedure efficiently under aseptic technique.

3.2 **Head Nurse** is responsible to make sure all staffs are aware of the procedure.

3.3 **Charge Nurse** is responsible for delegation.

3.4 **Registered Nurse** is responsible in assisting the physician during the procedure and taking are of post procedure activities.

4.0 POLICY

4.1. Only trained physician can insert the central venous catheter assisted by a registered nurse.

4.2. Entire procedure must be carried out under strict aseptic technique.

5.0 PROCEDURE (S)

5.1. ***Preparatory phase***

5.1.1. Identify the correct patient using IPSG goal 1.

5.1.2. Verify whether consent is complete.

5.1.3. Make sure patient kept for NPO at least 6 hours before insertion, if indicated.

5.1.4. Explain the procedure to the patient or parents.

5.1.5. Assemble all the necessary equipments as needed for insertion

5.1.5.1. Sterile set (CVP set)/cut down set

5.1.5.2. IV pole

5.1.5.3. Infusion solution

5.1.5.4. Infusion set with CVP manometer

5.1.5.5. Heparin flush system and pressure bag

5.1.5.6. Sterile dressing and tape

5.1.5.7. Appropriate PPE

5.1.5.8. Single transducer set

5.1.5.9. Arm board, if indicated

5.1.6. Hygienic hand washing. To prevent transfer of microorganisms.

5.1.7. Choose the site for insertion: jugular, femoral or subclavian vein.

5.1.8. Position the child in in supine position.

5.1.8.1. If to use the Jugular Vein, Place patient in trendelenburg's position with a small rolled towel placed under the shoulder (Subclavian Approach)

5.1.9. Turn the patients head away from the site.

5.1.10. Hygienic hand washing

5.1.11. Put on gown, cap, mask and sterile gloves both physician and assisting staff

5.1.12. Flush IV infusion set and manometer (measuring device) or prepare heparin flush for use with transducer. secure all connection to prevent to prevent air emboli and bleeding.

5.1.13. Attach the manometer to IV pole

5.1.14. The zero point of manometer should be on the level with patient right atrium

5.1.15. Calibrate /zero transducer and level port with patient right atrium

5.1.16. Institute the electro cardiogram monitoring.

5.2. ***Performance phase***

5.2.1. Clean the site using antiseptic lotions and help in draping the site.

5.2.2. Create a sterile environment and open the central line set.

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5.2.3. Assist the physician throughout the procedure by providing equipment.

5.2.4. Assist the patient in remaining motionless during insertion

5.2.5. Monitor for arrhythmias, Tachypnea, and tachycardia as catheter is threaded to great vein or right atrium

5.2.6. Once the physician confirms about the placement of catheter, connect the prime IV tubing or heparin flush system to catheter

5.2.7. Make sure no air bubbles in the tubing prior to connect to the catheter.

5.2.8. Aspirate and check for the back flow and allow the IV solution to flow.

5.2.9. The catheter is sutured in place.

5.2.10. Put sterile transparent dressing.

5.3. ***Follow up phase***

5.3.1. Label the dressing with Date and time of insertion.

5.3.2. Dispose all items according to infection control policy.

5.3.3. Wash hands thoroughly.

5.3.4. Take chest x-ray, to confirm the correct placement.

5.3.5. Record the nature of the procedure, type of suture material used, what type of lumen is used (Double or Triple lumen), as well as the patient response to procedure.

6.0 EQUIPMENTS

6.1. Sterile set (CVP set)/cut down set

6.2. IV pole

6.3. Infusion solution

6.4. Infusion set with CVP manometer

6.5. Heparin flush system and pressure bag

6.6. Sterile dressing and tape

6.7. Appropriate PPE

6.8. Single transducer set

6.9. Arm board, if indicated

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7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

8.1. MOH Guidelines for Pediatric Care

8.2. DPP-PICU-0007(V2) – Central Line Insertion – PMSH, 15/02/1442

8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) PICU standards, 3rd version, January 2016

9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer

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